

# High throughput gene perturbation and streamlined multimodal phenotyping in zebrafish

**Authors:** Abhinav Bachu\*, John Efromson\*, Claire Stockard, Randall Peterson, Saba Parvez

**Presenter:** Abhinav Bachu

Undergraduate Researcher  
Cell & Developmental Biology  
Feinberg School of Medicine

[abhinav.bachu@northwestern.edu](mailto:abhinav.bachu@northwestern.edu)

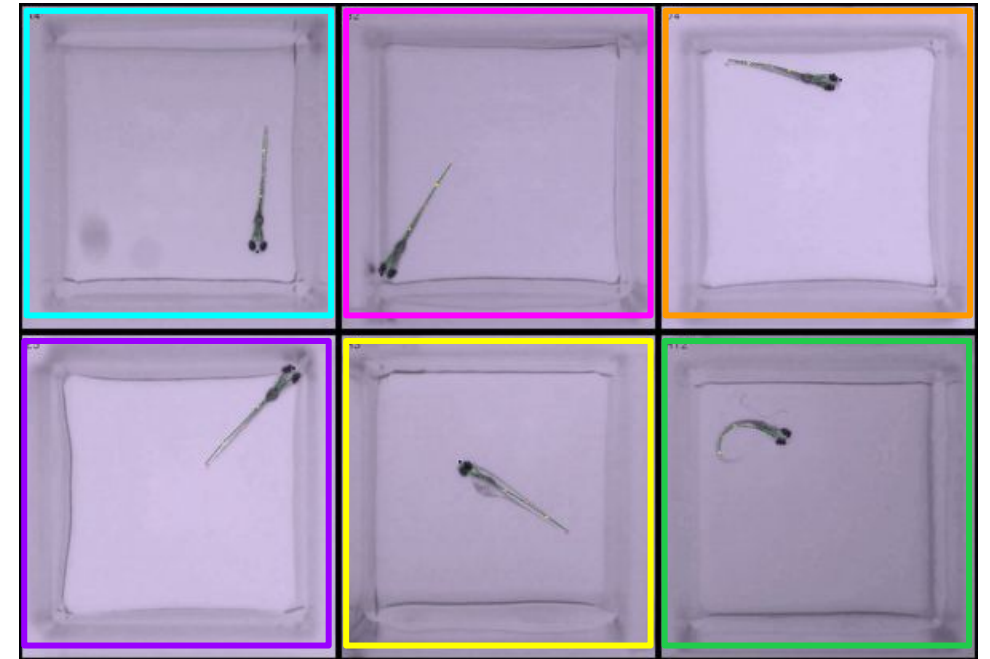


# Exigence

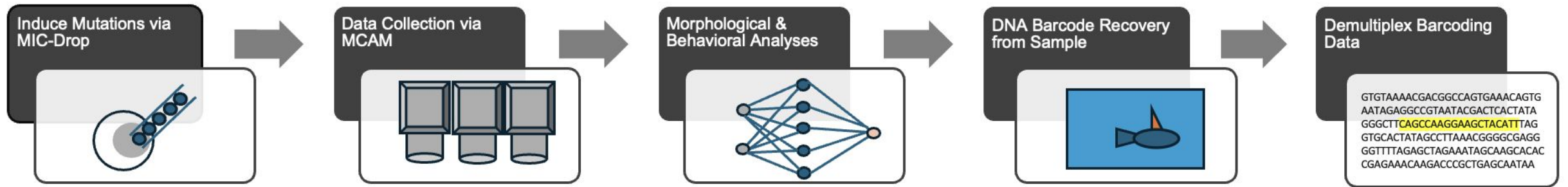
- CRISPR makes gene editing in zebrafish scalable — but phenotyping hasn't kept up
- Morphological and behavioral data are still collected manually or with low-throughput tools
- Most workflows don't link phenotype directly to genotype at scale
- **There's a need for a fast, standardized pipeline to connect gene edits to functional outcomes.**

# Introduction

- **Objective:** develop a scalable pipeline to connect genotype to phenotype in zebrafish
- Combine high-throughput CRISPR editing with automated imaging and analysis
- Define an easier and faster method to study how gene mutations affect morphology and behavior
- **This work brings together MIC-Drop, an automated high-throughput morphological and behavioral quantification system, and a plate-wide barcoding system.**

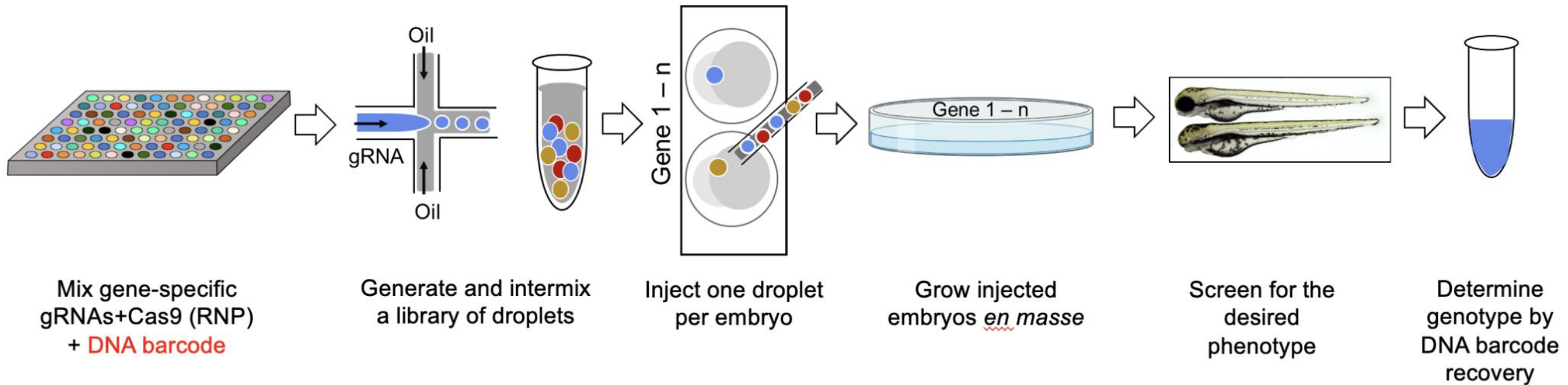


# Experimental Workflow



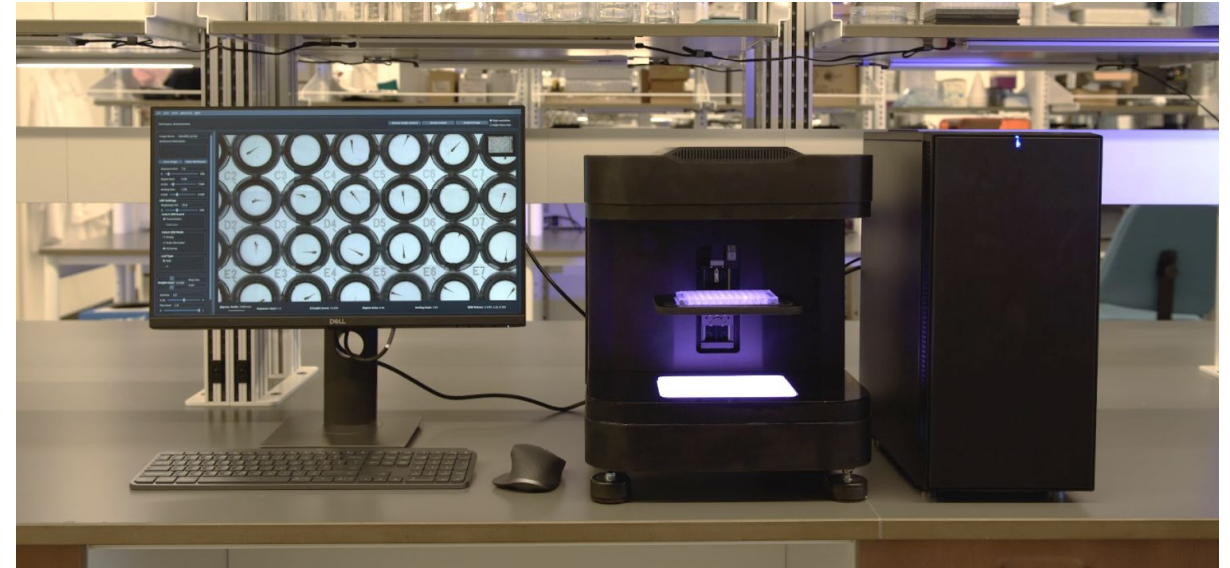
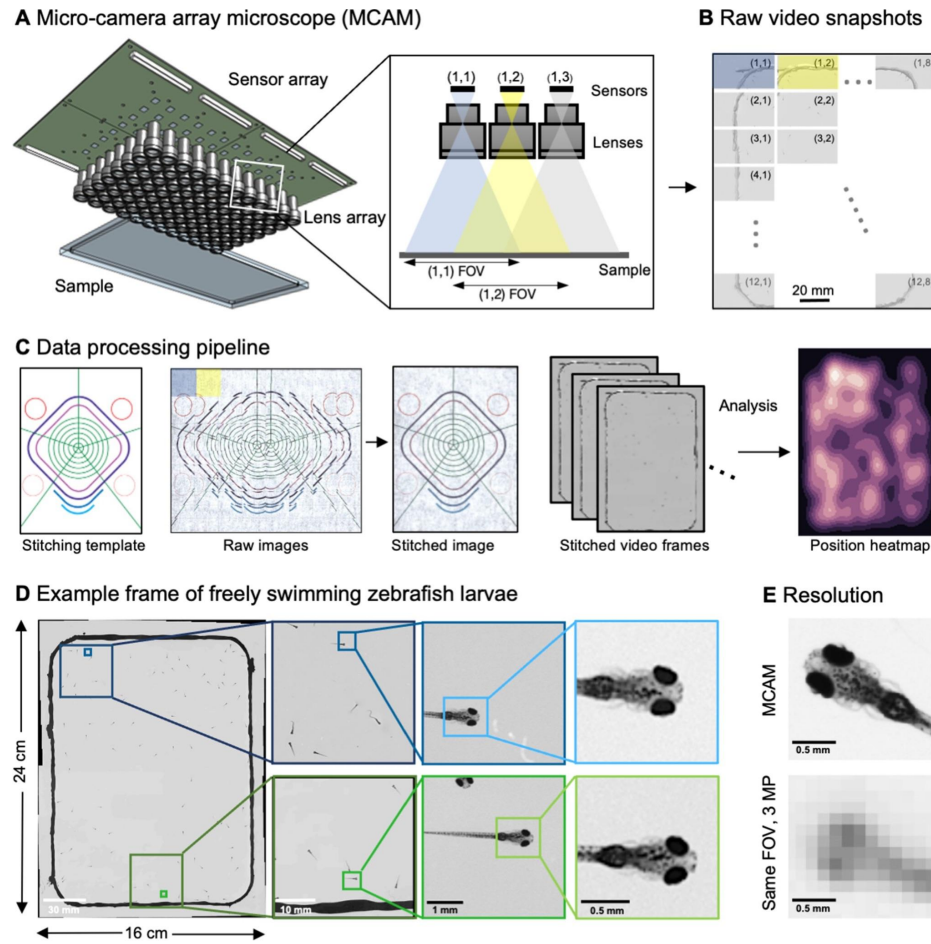
# MIC-Drop: Scalable Gene Perturbation

## MIC-Drop: Multiplexed Intermixed CRISPR Droplets



Parvez et al **2021** Science  
Parvez, Brandt, & Peterson **2023** Nature Protocols

# Imaging with MCAM

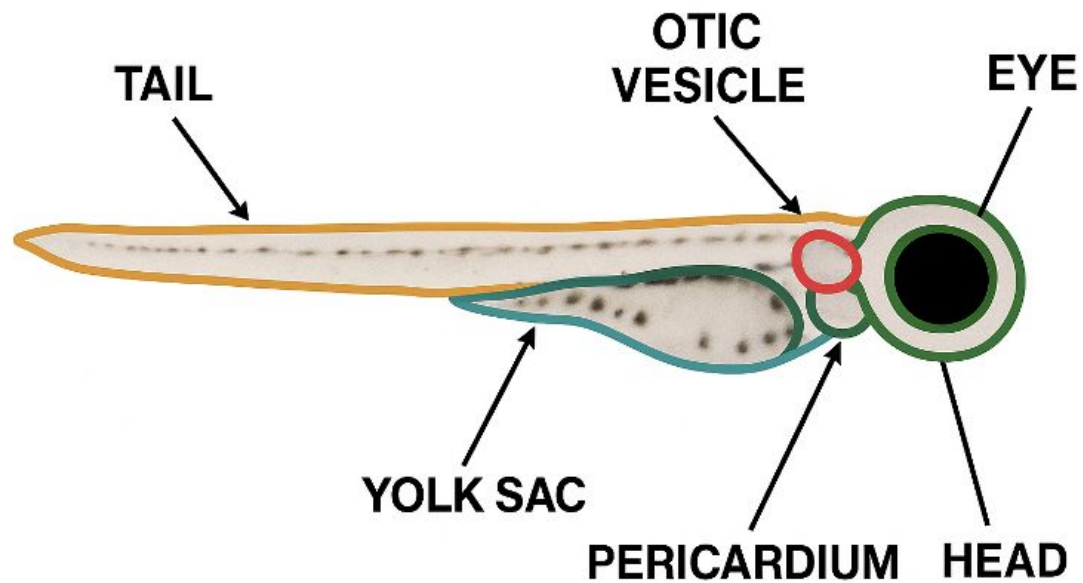


- Full 96-well Z-stack in <1 minute
- 0.9 gigapixels total, 3.2  $\mu\text{m}$  resolution
- Single system for all phenotypes

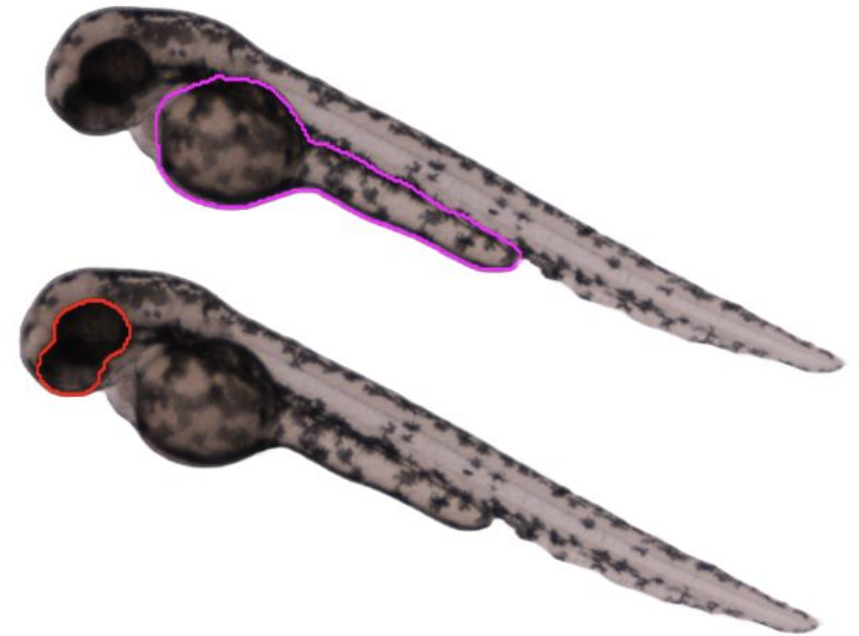
Thomson et al 2022 eLife



# CNN-Based Morphological Segmentation



UNet models trained on 7 regions



Outputs per-pixel masks with IoU evaluation

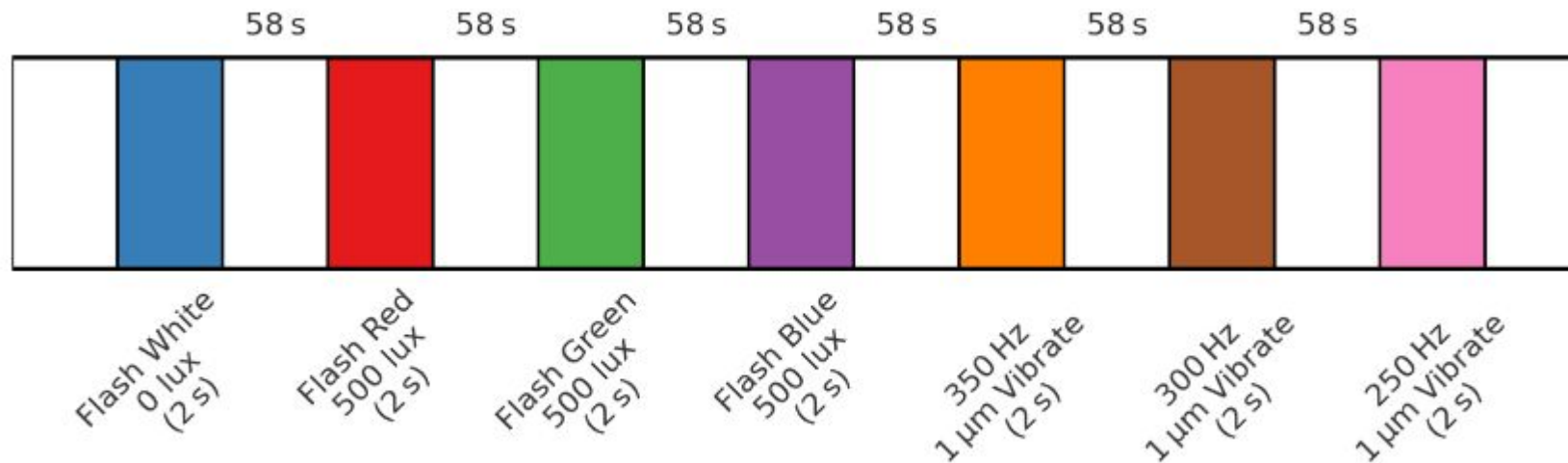
# Morphological Metric Outputs

metric outputs from mcam (wip)

- Extract tail length, head area, yolk diameter
- Plate-wide reproducibility and consistency

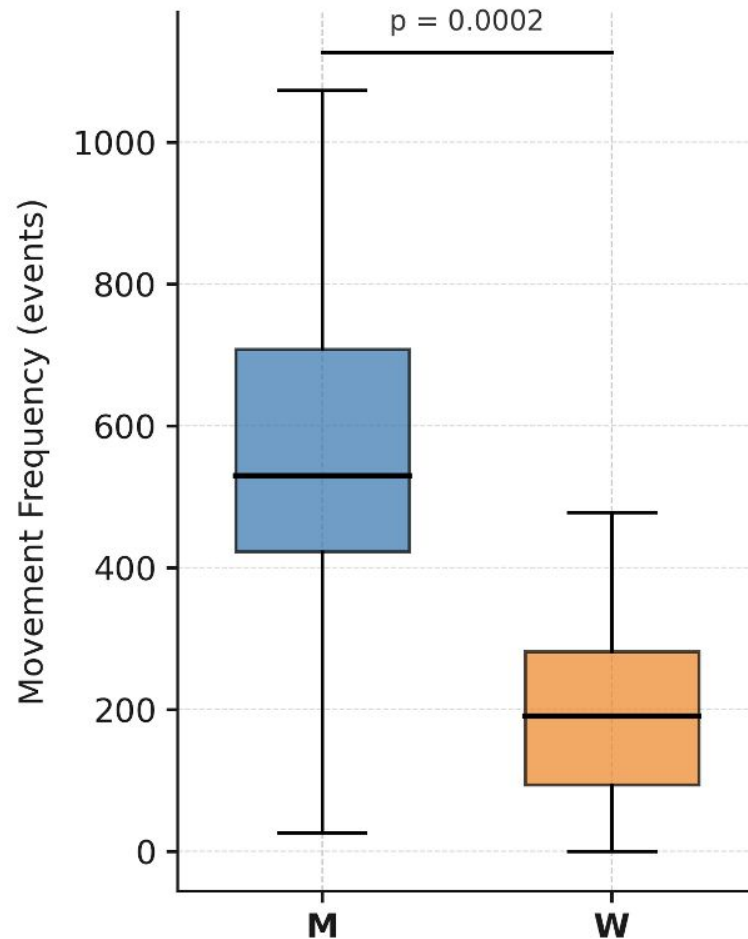


# Behavioral Assay Design



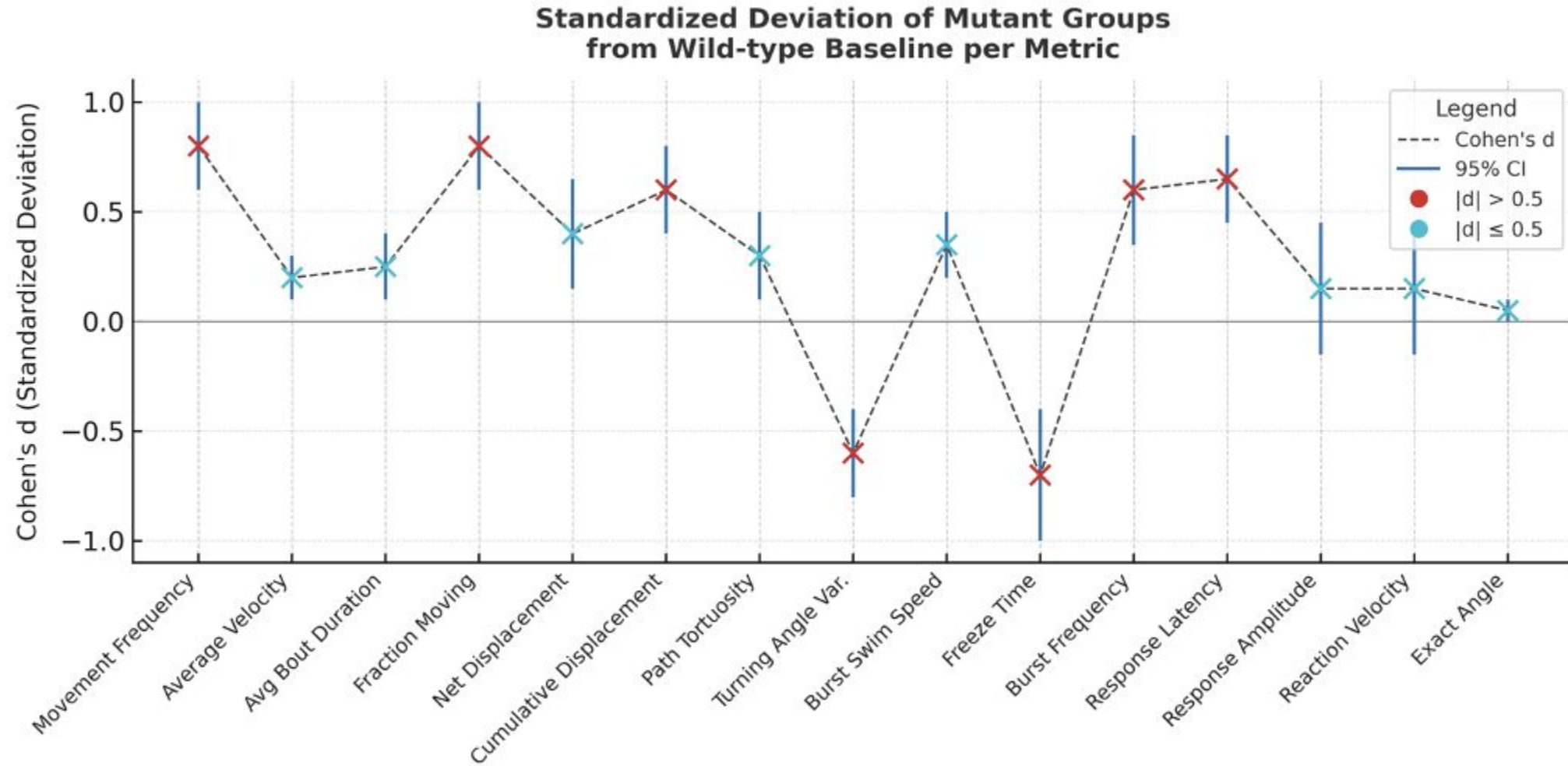
- Spontaneous and stimulus-based assays (flash, vibration)
- IR-based tracking + pose estimation

# Behavioral Metric Design



- Each metric captured separately
- Baseline and stimulus responses recorded for every larva
- Mutants compared to wild-type across all metrics
- Statistical significance assessed using t-tests with FDR correction

# Behavioral Fingerprinting



# Barcode Demultiplexing

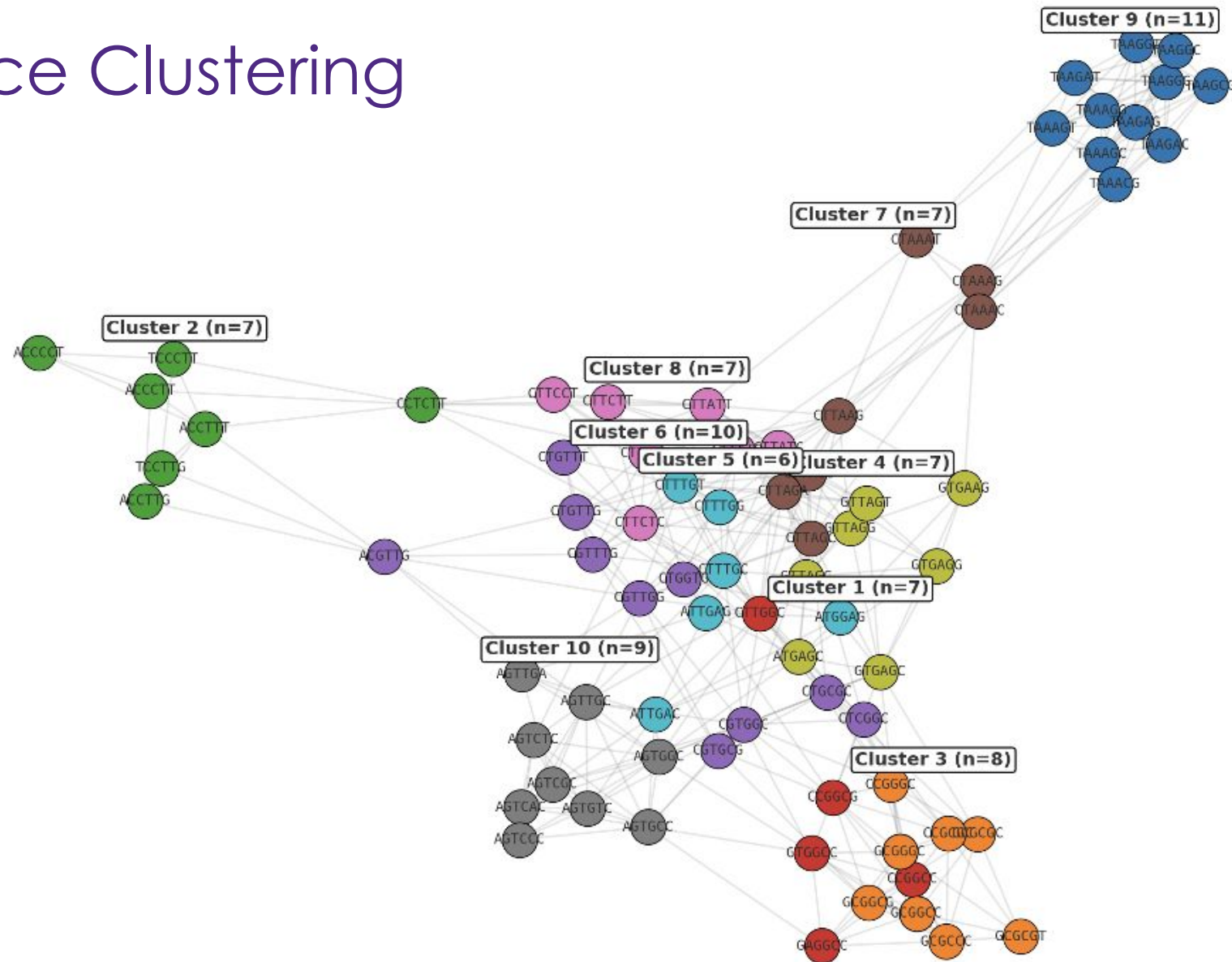
The screenshot shows a web application titled "96 Well Plate RNA Barcode Analysis". It features several input fields and buttons for data entry and file selection:

- Sequencing Data (.fastq or .txt):** A text input field with a "Browse" button to its right.
- Row Barcodes:** A large text input field.
- Column Barcodes:** A large text input field.
- Flanking Sequence 1:** A text input field.
- Flanking Sequence 2:** A text input field.
- Expected RNA Barcode Length:** A text input field.
- Edit Tolerance:** A text input field.
- Raw Data Output File (.txt):** A text input field with a "Browse" button to its right.
- Graphical Data Output File (.pdf):** A text input field with a "Browse" button to its right.
- Plate IDs (for multiple analysis):** A large text input field.

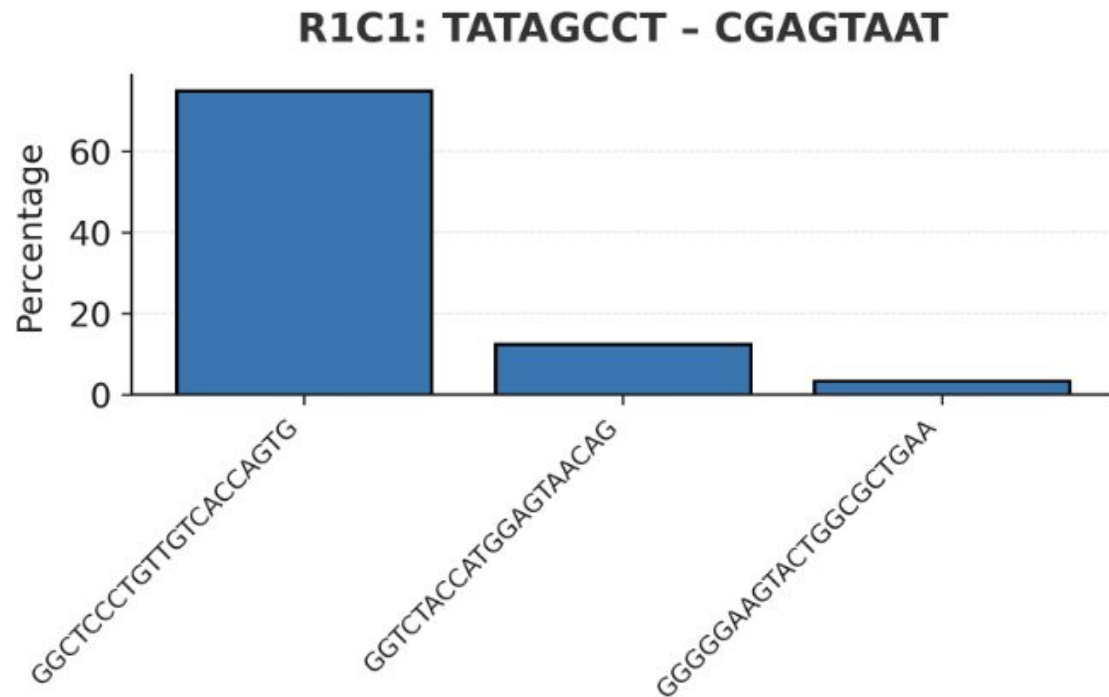
A progress bar is visible at the bottom of the interface.

- Takes in sequencing data (FASTQ) from post-phenotyping DNA
- Uses row/column IDs to assign each read to a specific well
- Outputs dominant genotype(s) for each larva in the 96-well plate
- Links phenotype data back to its gene edit origin

# Edit-Distance Clustering



# Genotype - Phenotype Mapping



|                                       |                                |                                |
|---------------------------------------|--------------------------------|--------------------------------|
| R1C1: GGCTCCCTGTTGTACCAGTG (75.00%)   | GGTCTACCATGGAGTAACAG (12.50%)  | GGGGAAGTACTGGCGTGAA (3.12%)    |
| R1C2: GGGCTTTCCGTAATATATAAC (75.44%)  | GGCAGCGAGTGCTGTGTCAG (5.26%)   | GGACTGACTTCGACCTGCAAA (5.26%)  |
| R1C3: GGCGGAGGCATCATCGTCTCC (72.83%)  | GGCCCCGCTGGAATGAGGTGTA (3.26%) | GGCACCAGTTCTACCCGACTG (3.26%)  |
| R1C4: GGGGAAGTACTGGCGTGAA (58.49%)    | GGACTGACTTCGACCTGCAAA (9.43%)  | GGCAGGTAAGTTATGCAACT (9.43%)   |
| R1C5: GGCCAGGTAAAGTTATGCAACT (69.37%) | GGCGGAGGCATCATCGTCTCC (6.31%)  | GGTCTGCTGTGGCAGCCGT (5.41%)    |
| R1C6: GGCGGCATCAGCTCGTGTCTG (46.67%)  | GGGGTCAGAATTACAACGACA (20.00%) | GGACTGACTTCGACCTGCAAA (13.33%) |
| R1C7: GGACTGACTTCGACCTGCAAA (64.80%)  | GGCGGCATCAGCTCGTGTCTG (14.40%) | GGAGTCTCTCTCATCGCTCG (4.00%)   |
| R1C8: GGACTCTGAGCGGCTTCCAGG (58.16%)  | GGAGATGACTGTCCATTGGTC (11.22%) | GGATGAGCCATCCATGAATGC (4.08%)  |
| R1C9: GGACCCAGTTCTACCCGACTG (63.64%)  | GGCGACAGGCACCGAACAGAG (12.73%) | GGGGAGAGACTGTGTTTACC (5.45%)   |
| R1C10: GGGGAGAGACTCTGTTTACCC (79.57%) | GGTTTAGACGTAAAGAGTTAC (7.53%)  | GGACTGACTTCGACCTGCAAA (2.15%)  |
| R1C11: GGAAGAACCAGGAGGCGCGG (60.38%)  | GGGAGTCACGCAGTGTGTGG (13.21%)  | GGACTCTGAGCGGCTTCCAGG (3.77%)  |
| R1C12: GGAGGACGTGAGGAACTGG (62.26%)   | GGTCCGTGTAGTCCAGACTGA (5.66%)  | GGCAACTCAAGCTCACGGTG (3.77%)   |
| R2C1: GGATTGACCGGACCTGCGAG (64.00%)   | GGGGCTGGAGTGCAACTATG (6.67%)   | GGAAGTGGCGGTATTGCCCA (2.67%)   |
| R2C2: GGCTCCCATCACCTGACAGT (68.14%)   | GGATTGACCGGACCTGCGAG (5.31%)   | GGACTGACTTCGACCTGCAAA (3.54%)  |
| R2C3: GGGGTGAGTAATACAACGACA (77.69%)  | GGACTGACTTCGACCTGCAAA (4.62%)  | GGCCGAGATACATAATGGATG (2.31%)  |
| R2C4: GGCCCTTAAACGGGCGAGGG (63.33%)   | GGCTGTGCGATCCGTATGTAC (18.33%) | GGCTCTCTGTGGACAGGGCC (3.33%)   |
| R2C5: GGAAGTCTGAGCGCTCGGGT (56.86%)   | GGGGCTGGAGTGCAACTATG (9.80%)   | GGCACGGTAAGTTATGCAACT (4.90%)  |
| R2C6: GGACTGACTTCGACCTGCAAA (69.23%)  | GGCGGCATCAGCTCGTGTCTG (11.54%) | GGGCAAGAACCCTTACCAG (3.85%)    |
| R2C7: GGCGGAGGCATCATCGTCTCC (62.99%)  | GGTCTTGTGTGGCAGCCGT (9.74%)    | GGACTGACTTCGACCTGCAAA (3.90%)  |
| R2C8: GGGCAAGAACCTCTACCGAG (66.96%)   | GGATTAATGCACCTTTCCAG (13.39%)  | GGACTCTGAGCGGCTTCCAGG (2.68%)  |
| R2C9: GGCCCAAGAACCTCTACCGAG (70.97%)  | GGGGAGAGACTCTGTTTACC (8.60%)   | GGAAATGGCTACCTCTCTCG (3.23%)   |
| R2C10: GGGGAGAGACTCTGTTTACC (68.09%)  | GGGCATGCTGAGCGAGCAGGA (11.70%) | GGACTGACTTCGACCTGCAAA (2.13%)  |
| R2C11: GGTGCGAGTACAGGCGCTGG (79.17%)  | GGTCCCGGTCTCCGTTCGGC (2.08%)   | GGCAGCGAGTGCTTGTTCAG (2.08%)   |
| R2C12: GGCTCTCTGTGGACAGGGCC (77.32%)  | GGTACGTTACCATGCAAGTC (2.06%)   | GGTCCGTGTAGTCCAGACTGA (2.06%)  |
| R3C1: GGGGAAGTACTGGCGTGAA (31.82%)    | GGCTGTGCGATCCGTATGTAC (22.73%) | GGCACGGTAAGTTATGCAACT (6.82%)  |
| R3C2: GGCTCCCATCACCTGACAGT (72.60%)   | GGGTTCCACGAAAGACCTCG (4.11%)   | GGACTGACTTCGACCTGCAAA (2.74%)  |
| R3C3: GGCGAGGAGTGCTGTGTCAG (79.07%)   | GGCCATCCTTACCTACAGACA (8.53%)  | GGGCCAGGCTCATCTGATAGG (1.55%)  |
| R3C4: GGGGTCCAGCCTTCATATACG (69.79%)  | GGACCATGTGTACACCGTAG (5.21%)   | GGCCCGCTGGAATGAGGTGA (5.21%)   |
| R3C5: GGACGCTCTGACATATCTCGA (70.50%)  | GGAAGTCTATCGTTCTCGATA (10.79%) | GGTCCGTGTAGTCCAGACTGA (2.88%)  |
| R3C6: GGCTCCCATCACCTGACAGT (63.19%)   | GGCGGAGGCATCATCGTCTCC (15.97%) | GGCAGCCTCTGACTATCTCGA (2.78%)  |
| R3C7: GGCGGAGGCATCATCGTCTCC (54.90%)  | GGCTCCATCACCTGACAGT (9.80%)    | GGTAGCCTCTGACTATCTCGA (5.88%)  |
| R3C8: GGATGAGCCATCCATGAATGC (67.69%)  | GGTCCGTGTAGTCCAGACTGA (3.85%)  | GGCTCCCATCACCTGACAGT (2.31%)   |
| R3C9: GGAAGAACCAGGAGGCGCGG (57.38%)   | GGTTTGTCAACAACTTACCG (18.03%)  | GGGGAGAGACTCTGTTTACC (8.20%)   |
| R3C10: GGGGAGAGACTCTGTTTACC (66.07%)  | GGCAGCCCTTGGCAGGTGTT (5.36%)   | GGGAACCCGACCTCCGTATC (3.57%)   |
| R3C11: GGGCCAGGCTCATCTGATAGG (59.57%) | GGTTTGTCAACAACTTACCG (7.45%)   | GGTCCGTGTAGTCCAGACTGA (2.13%)  |
| R3C12: GGTCCGTGTAGTCCAGACTGA (67.24%) | GGAGGCTGTCCCTCCTGAG (12.93%)   | GGATGAGCCATCCATGAATGC (2.59%)  |



# Discussion

- Our platform allows **high-throughput, standardized phenotyping** of zebrafish mutants
- By automating imaging, segmentation, behavior tracking, and genotype assignment, we **eliminate major bottlenecks** in reverse genetics
- Results **matched manual annotations**, validating the accuracy of the system
- This approach **compares favorably** to existing tools like DanioVision, VAST, and Zebrabox, with improved scalability and data integration
- The system is **modular, cost-effective, and adaptable for broader use** in developmental biology and disease modeling

# Acknowledgements

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# Questions?

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